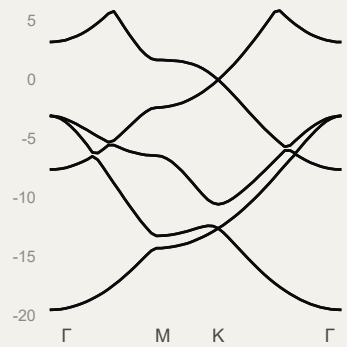
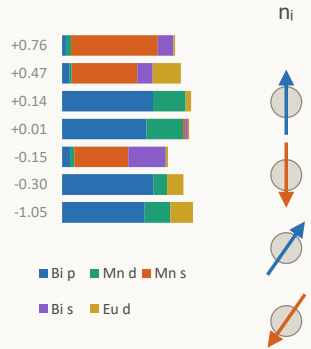
(a) DFT output



Quantum ESPRESSO or VASP

(b) Read



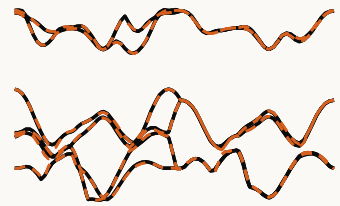
Electron Count, Orbital Projections, Moment Direction of every site

(c) Model



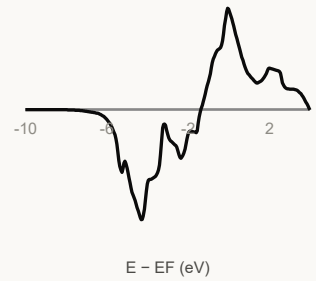
Orbitals or Symmetry Groups, Group-pair integrals, Shells, SOC λ L·S, Exchange Δ $n_i \cdot \sigma$

(d) Fit



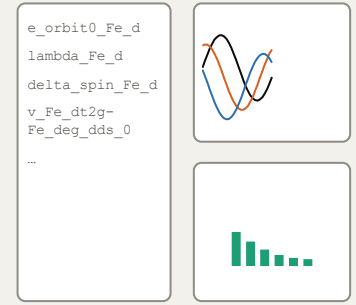
Weighted least squares with warm starts

(e) Berry phase



Berry curvature, AHC, SHC, Wilson loops, Chern numbers from the fitted model

(f) Outputs



Parameter file, Band comparison, Berry Phase based properties

SKFitter Functions